

Let  $V$  be a fin. dim vector space, and  $T: V \rightarrow V$  be a linear map.  
Define  $\chi_T(t) = \det(tI - [T]_B)$  be the characteristic polynomial,  
 $m_T(t)$  be the minimal polynomial.

Lemma. For  $c \in F$ ,

$$\chi_T(c) = 0 \text{ if and only if } \ker(T - cI) \neq \{0\}.$$

Proof.  $\chi_T(c) = 0 \Leftrightarrow \det(cI - [T]_B) = 0 \Leftrightarrow cI - [T]_B$  is non-singular.

Lemma. For  $c \in F$ ,

$$\chi_T(c) = 0 \text{ if and only if } m_T(c) = 0.$$

Proof. Suppose that  $\chi_T(c) = 0$ . Then,  $T(\alpha) = c \cdot \alpha$  for some non-zero  $\alpha \in V$ .  
Therefore,  $0 = m_T(T)(\alpha) = m_T(c) \cdot \alpha$ , but since  $\alpha \neq 0$ ,  $m_T(c) = 0$ .

Conversely, suppose that  $m_T(c) = 0$ . Then,

$$m_T(t) = (t - c) \cdot g(t) \text{ for some } g(t) \in F[t].$$

By the minimality of  $m_T$ ,  $g(T) \neq 0$ , so that there exists  $\alpha \in V$  such that  $g(T)(\alpha) \neq 0$ , thus

$$(T - cI)(g(T)(\alpha)) = m_T(T)(\alpha) = 0$$

So proved.  $\square$

The Primary decomposition theorem.

Let  $T$  be a linear operator on the finite-dimensional vector space  $V$  over a field  $F$ .  
Let  $P$  be the minimal polynomial for  $T$  with the irreducible decomposition

$$P = p_1^{r_1} \cdot p_2^{r_2} \cdots p_k^{r_k} = \prod_{i=1}^k p_i(x)^{r_i}.$$

Then,  
 $V = \bigoplus_{i=1}^k \ker p_i^{r_i}(T)$  and each  $\ker p_i^{r_i}(T)$  is invariant  $T$ .

Moreover, the restriction of  $T$  by  $\ker p_i^{r_i}(T)$  has the minimal polynomial as  $p_i^{r_i}$ .

Thm. Let  $\alpha$  be any non-zero vector in  $V$  and let  $p_\alpha$  be the  $T$ -annihilator of  $\alpha$ .

- 1)  $\deg p_\alpha = \dim Z(\alpha, T)$
- 2)  $Z(\alpha, T) = \text{span} \{T^k \alpha \mid 0 \leq k \leq \deg p_\alpha\}$
- 3) The minimal polynomial of  $T|_{Z(\alpha, T)}$  is  $p_\alpha$ .

Cyclic decomposition Theorem.

Thm. Let  $T: V \rightarrow V$  be a linear operator on finite-dimensional space  $V$ .

Suppose that the minimal polynomial of  $T$  has a irreducible decomposition

$$m_T(x) = p_1(x)^{r_1} \cdots p_k(x)^{r_k}$$

Then, by the Primary decomposition,  $V$  is decomposed by  $T$ -invariant spaces

$$V = \ker p_1^{r_1} \oplus \cdots \oplus \ker p_k^{r_k}$$

With  $T|_{\ker p_i^{r_i}}$  has minimal polynomial as  $p_i^{r_i}$ .

Now, for fixed  $1 \leq i \leq k$ , denote  $T_i = T|_{\ker p_i^{r_i}}$  and  $W_i = \ker p_i^{r_i}$ .

Then, there are non-zero vectors  $\alpha_1, \dots, \alpha_n$  such that

$$W_i = Z(\alpha_1, T_i) \oplus \cdots \oplus Z(\alpha_n, T_i)$$

satisfying for some unique integers  $1 \leq r_n \leq \cdots \leq r_2 \leq r_1 = r_i$ ,

$$m_{T_i|_{Z(\alpha_s, T_i)}} = p_i^{r_s}.$$

Primary decomposition: splits case.

Suppose that the minimal polynomial of  $T$  splits as

$$m_T(t) = (t - \lambda_1)^{r_1} \cdots (t - \lambda_k)^{r_k}.$$

Then, by the Primary decomposition,

$$V = \ker(T - \lambda_1)^{r_1} \oplus \cdots \oplus \ker(T - \lambda_k)^{r_k}.$$

Now, we get the resolution of identity

$$T = T \circ E_1 + \cdots + T \circ E_k \quad \text{where } E_i \text{ is projection onto } \ker(T - \lambda_i)^{r_i} \\ \text{along } \bigoplus_{j \neq i} \ker(T - \lambda_j)^{r_j}.$$

and let

$$D = \lambda_1 E_1 + \cdots + \lambda_k E_k.$$

Then,

$$N = T - D = (T - \lambda_1 I) \circ E_1 + \cdots + (T - \lambda_k I) \circ E_k.$$

is nilpotent, with the order is at most  $\max\{r_1, \dots, r_k\}$ .

Moreover, for  $i \neq j$ ,

$$(T - \lambda_i I) \Big|_{\ker(T - \lambda_j)^{r_j}} = \underbrace{(\lambda_j - \lambda_i) \cdot I}_{\text{invertible}} + \underbrace{(T - \lambda_j I) \Big|_{\ker(T - \lambda_j)^{r_j}}}_{\text{nilpotent}}$$

So it is invertible.

Jordan form.

Let  $T: V \rightarrow V$  be a linear operator on  $n$ -dimensional vector space over  $F$ , with  $\chi_T(t) = (t - \lambda_1)^{r_1} \cdots (t - \lambda_k)^{r_k}$

and  $m_T(t) = (t - \lambda_1)^{s_1} \cdots (t - \lambda_k)^{s_k}$  ( $1 \leq s_i \leq r_i, \forall i$ .)

1).  $\bigcup_{N=1}^{\infty} \ker(T - \lambda_i)^N = \ker(T - \lambda_i)^{s_i}$ .

Proof. The  $\supseteq$  inclusion is clear. And, by the Primary decomposition, given  $x \in \bigcup_{N=1}^{\infty} \ker(T - \lambda_i)^N$ , there is unique representation

$$x = x_1 + \cdots + x_k \quad \text{where } x_s \in \ker(T - \lambda_s)^{s_s}, \forall s.$$

Fix  $m \in \mathbb{N}$  such that  $x \in \ker(T - \lambda_i)^m$ . That is,

$$(T - \lambda_i I)^m(x) = \sum_{s=1}^k (T - \lambda_i I)^m(x_s) = 0.$$

Meanwhile, each  $W_s$  is  $T - \lambda_i I$  invariant and  $(T - \lambda_i I)|_{\ker(T - \lambda_s I)^{s_s}}$  is invertible, therefore  $x_s = 0$  for all  $s \neq i$ . Finally,  $x = x_i \in \ker(T - \lambda_i)^{s_i}$ . So proved.  $\square$

$\exists x_1, x_2, \dots, x_{s_i}$  not independent.

2).  $\dim \ker(T - \lambda_i)^{s_i} = \dim \bigcup_{N=1}^{\infty} \ker(T - \lambda_i)^N = r_i$ .

Proof. By the Primary decomposition,  $V = \bigoplus \ker(T - \lambda_i)^{s_i}$ , so that

$$\chi_T(t) = \prod_{i=1}^k \chi_{T|_{\ker(T - \lambda_i)^{s_i}}}(t) = \prod_{i=1}^k (t - \lambda_i)^{\dim \ker(T - \lambda_i)^{s_i}}, \quad \text{So proved. } \square$$

In  $\ker(T - \lambda_i)^{s_i}$ , it has only  $\lambda_i$  as eigenvalue.

3).  $s_i =$  largest Jordan block size for  $\lambda_i$ .

$\dim \ker(T - \lambda_i I) =$  number of  $\lambda_i$ -blocks.

Proof.  $J_{n_1}(\lambda) \oplus \cdots \oplus J_{n_g}(\lambda)$ , then  $m_{T|_V}(t) = \text{lcm}((t - \lambda)^{n_1}, \dots, (t - \lambda)^{n_g}) = (t - \lambda)^{\max n_i}$ .

Summary.

Jordan form.

Let  $T: V \rightarrow V$  be a linear operator on  $n$ -dimensional vector space over  $F$ ,

with  $\chi_T(t) = (t - \lambda_1)^{r_1} \cdots (t - \lambda_k)^{r_k}$

and  $m_T(t) = (t - \lambda_1)^{s_1} \cdots (t - \lambda_k)^{s_k} \quad (1 \leq s_i \leq r_i, \forall i.)$

Then, the Jordan form is:  $s_{i1} \geq s_{i2} \geq s_{i3} \geq \dots$ ,

$$J = \begin{bmatrix} J_{s_1}(\lambda_1) & & & \\ & J_{s_{i2}}(\lambda_1) & & \\ & & \ddots & \\ & & & J_{s_{i m_i}}(\lambda_1) & \\ & & & & J_{s_2}(\lambda_2) & \ddots \end{bmatrix}$$

$r_1$

Computations.

$\mathbb{H}K$ ,  $1, 3, 3$ .

Let  $A \in M_{5 \times 5}(\mathbb{C})$  with

$$\chi_A(t) = (t-2)^3 \cdot (t+1)^2 \quad \text{and} \quad m_A(t) = (t-2)^2 \cdot (t+1).$$

Determine the Jordan form of  $A$ .

Proof. Since  $(t+1)$  has multiplicity 2 in  $\chi_A$  and 1 in  $m_A$ , the Jordan form of  $A$  has  $[-1] \oplus [-1]$  as a block corresponding to  $-1$ , and  $(t-2)$  has multiplicity 2 in  $m_A$ , it has a Jordan block

$$J_2(2) = \begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}$$

and  $(t-2)$  has multiplicity 1 in  $\chi_A$ , the last block is forced  $[2]$ . Consequently, the Jordan form is

$$J_A = \begin{bmatrix} 2 & 1 & & & \\ & 2 & & & \\ & & 2 & & \\ & & & -1 & \\ & & & & -1 \end{bmatrix}$$

TKAIST, 2027 spring.

Let  $A \in M_{8 \times 8}(\mathbb{R})$ . Suppose  $\chi_A(x) = (x+1)^5 \cdot (x-2)^3$

1). Assume

$\dim \ker(A+I) = 2$ ,  $\dim \ker(A+I)^2 = 4$ ,  $\dim \ker(A+I)^3 = 5$   
 $\dim \ker(A-2I) = 1$ ,  $\dim \ker(A-2I)^2 = 2$ ,  $\dim \ker(A-2I)^3 = 3$ .  
Determine all possible Jordan Canonical form of  $A$ .

2). Assume

$$m_A(x) = (x+1)^3 \cdot (x-2)^3.$$

Determine all possible Jordan Canonical form of  $A$ .

Proof. 1). Since  $\dim \ker(A+I)^3 = 5$  equals to the multiplicity of  $x+1$  in  $\chi_A$ ,

$$\mathbb{R}^8 = \ker(A+I)^3 \oplus \ker(A-2I)^3. \text{ and, moreover,}$$

$\dim \ker(A+I) = 2$  implies  $A$  has two Jordan blocks for  $-1$ ,

$\dim \ker(A-2I) = 1$  implies  $A$  has only one Jordan block for  $2$ .

Moreover,  $\dim \ker(A+I)^2 - \dim \ker(A+I) = 2$  but

$\dim \ker(A+I)^3 - \dim \ker(A+I)^2 = 1$  therefore it has Jordan blocks of size 2 and 3. Finally, there is only one possible form

$$J_A = J_3(-1) \oplus J_2(-1) \oplus J_3(2).$$

2). Since  $(x+1)$  has multiplicity 3 in  $m_A$ , there are possible partitions

$$5 = 3 + 2 = 3 + 1 + 1,$$

and since  $x-2$  has multiplicity 3 in  $m_A$ , there are two possible forms:

$$J_A = J_3(-1) \oplus J_2(-1) \oplus J_3(2) \text{ or}$$

$$= J_3(-1) \oplus [-1] \oplus [-1] \oplus J_3(2)$$

$\square$

HK, 7.3, 15 ~ 16.

Let  $N \in M_{\text{non}}(\mathbb{C})$  be a nilpotent matrix over Complex numbers.

For any non-zero  $c \in \mathbb{C}$ ,  $cI + N$  has a square root.

Moreover, deduce that every invertible matrix over  $\mathbb{C}$  has a square root.

Proof. Let  $A = c \cdot \left( \sum_{k=0}^{n-1} \frac{1}{2^k} \cdot N^k \right)$ .